SAHIL RATHEE

SENIOR SYSTEMS ARCHITECT & DISTRIBUTED INFRASTRUCTURE ENGINEER

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EXECUTIVE SUMMARY

Senior Systems Architect and Backend Infrastructure Engineer with 4+ years of production experience architecting high-concurrency distributed systems, low-latency financial event buses, and autonomous agentic workflows. Proven track record migrating legacy monoliths to declarative zero-drift cloud environments, shrinking application payloads by 80%, reducing AI operational expenditure by 70% via AST context optimization, and engineering sub-millisecond execution pipelines.

TECHNICAL COMPETENCIES

Languages: Go (Golang), Python, TypeScript, C++, Dart, Kotlin, SQL, Gremlin, Bash
Frameworks & Mobile: Next.js (SSR/Edge), React Native (Expo), Flutter, Node.js, FastAPI, Cocos Creator Engine
Cloud & Infrastructure: Kubernetes (EKS), Docker, Terraform IaC, Ansible, Jenkins CI/CD, AWS (Neptune, OpenSearch), GCP, Cloudflare
Databases & Streaming: PostgreSQL, MongoDB, Redis, Qdrant Vector DB, Amazon Neptune (Graph), NATS JetStream, RabbitMQ
Core Systems & Domains: High-Frequency Telemetry, Double-Entry Ledgers, Multi-Agent Swarms, RAG Optimization, Monte Carlo Simulation

PROFESSIONAL EXPERIENCE

Full Stack Software Engineer — NaiBooking

Aug 2026 – Present • Full-time • Remote (Bangalore, India)

- Architected full-stack salon & barber marketplace: web portal and admin console in **React / Next.js**, client and business mobile apps in **React Native (Expo)** live on Google Play & App Store.
- Engineered Node.js microservices leveraging **MongoDB** and **PostgreSQL** with transaction ledgers, advisory locking, and multi-tier Redis caching to handle high-concurrency booking spikes.
- Automated AWS cloud infrastructure via **Terraform IaC**; integrated Razorpay automated subscription billing for salon CRM with an architectural roadmap for Stripe global expansion (UAE, USA, Singapore, Australia).

Tech Stack: *TypeScript, React, Next.js, React Native (Expo), Node.js, MongoDB, PostgreSQL, Redis, AWS, Terraform IaC, Docker, CI/CD*

Lead Software Engineer — Orealfit

May 2026 – Jul 2026 • Full-time Contract • On-site (Sector 44, Gurugram, India)

- Contractually architected an enterprise sports ecosystem featuring **real-time scorecard streaming** and interactive advertising modules, elevating user retention and backend throughput.
- Transitioned engineering operations to **autonomous agentic workflows** by deploying autonomous dev and test agent pipelines, accelerating end-to-end feature delivery velocity.
- Optimized RAG pipeline efficiency and AI infrastructure through custom token telemetry and prefix caching, achieving a **70% reduction in monthly AI token expenditure**.
- Engineered a high-performance wellness e-commerce platform using monorepo architecture and an edge-computed billing/GST solution with sub-50ms latency.

Tech Stack: *Flutter, Next.js, Go, C++, Node.js, PostgreSQL, Redis, Qdrant Vector DB, Supabase, Cloudflare Workers, Docker*

Quantitative Research & Algorithmic Trading — Independent Deep-Dive

Dec 2025 – Apr 2026 • Self-Directed

- Dedicated engineering deep-dive into **Quantitative Modeling, Algorithmic Execution, and High-Frequency Trading (HFT)** architectures.
- Formulated mathematical trading models, statistical arbitrage algorithms, and tick-level backtesting engines across equities and foreign exchange.
- Engineered sub-millisecond in-memory ring buffers and MetaTrader 5 execution bridges streaming real-time order-book telemetry via NATS JetStream (Zenith & Mitochondria).
- Developed invariant-based risk management safeguards including dynamic position sizing, trailing circuit breakers, and automated drawdown enforcement.

Focus & Domains: *Quantitative Research, Algorithmic Trading, High-Frequency Trading (HFT), MetaTrader 5, Python, C++, Go, NATS JetStream*

Solution Architect — Gym Wise India Private Limited

Jul 2024 – Nov 2025 • Full-time • On-site (Sector 14, Gurugram, India)

- Re-architected a legacy monolithic codebase into **3 standalone cross-platform Flutter/Dart apps** via build flavors: **GymWise Customer**, **GymWise Partner** (with biometric attendance), and **GymWise Admin**, alongside a Next.js web portal, live on Google Play & App Store.
- Re-engineered backend microservices from PHP Laravel to **Go (Golang)** backed by a Redis buffer and **Qdrant Vector DB** for member search and similarity matching.
- Built multi-cloud infrastructure: **AWS** compute and container orchestration managed via **Terraform IaC and Ansible**; **GCP** for Maps APIs and OAuth services.
- Implemented gym marketplace business logic: zero upfront cost to gym owners, automated **30% commission** collection on customer bookings, and algorithmic dynamic pricing based on user engagement and demand.

Tech Stack: *Flutter, Dart, Go (Golang), AWS, GCP, Redis, Qdrant Vector DB, PostgreSQL, Terraform IaC, Ansible, Docker Swarm, Play Console*

AI Research & Agentic Systems — Independent Deep-Dive

Apr 2024 – Jun 2024 • Self-Directed

- Dedicated research deep-dive between BigHit and GymWise into **Autonomous AI Agents and Multi-Agent Workflows**.
- Researched and built AST context compression algorithms, dynamic prompt caching, and isolated Docker execution environments for autonomous coding agents (Meeseeks framework).
- Designed hybrid agentic RAG retrieval architectures, hierarchical vector indexing strategies, and automated decision verification loops.

Focus & Domains: *Autonomous AI Agents, Multi-Agent Swarms, AST Code Parsing, Vector Embeddings, LLM Orchestration, Python, Docker*

Full Stack Engineer — Bighit Nov 2022 – Mar 2024 • Full-time • On-site (Andheri West, Mumbai, India)

- Developed real-time **Socket.io Node.js server** powering dynamic chat for **2,000 Concurrent Active Users (CCU)** (~400 events/sec peak throughput).
- Built real-time score updates and live telemetry engine for **1,000 CCU** live-streamed football matches (U-14 tournaments, BigHit Sportz), featuring a dedicated third-umpire web console with access control and abstracted scoring workflows.
- Migrated legacy relational SQL database to **Amazon Neptune graph DB (Gremlin)**, optimizing multi-degree athlete-to-scout social graph traversal and talent discovery queries.
- Modernized mobile & backend infrastructure (React Native & AWS IaC), achieving an **~80% reduction in app bundle size** and an **~60% increase in server efficiency**.
- Integrated and managed AWS cloud infrastructure with **Amazon Neptune** graph DB and **OpenSearch** clusters, reducing API response latency by **80%**.
- Led and mentored a team of 6 engineers: ran technical roadmaps, sprint planning, code reviews, and developer onboarding.

Tech Stack: *TypeScript, React Native, Node.js, Socket.io, GraphQL, AWS Neptune (Graph DB), OpenSearch, Docker, Kubernetes, AWS IaC, CI/CD*

Full Stack Developer — Gamix Labs Mar 2022 – Oct 2022 • Full-time • On-site (Noida, India)

- Engineered real-time mathematical simulation engines for Vegas slot machines (HTML5/Facebook Instant Games on Cocos Creator Engine) with Node.js and RabbitMQ event streaming backends.
- Formulated and verified Return-to-Player (RTP) and win frequency probability algorithms via Monte Carlo simulations (**10,000,000+ spins at 99.9% confidence interval** across win bands: 65.2% zero-win, 15.4% <1x, 12.1% 1-5x, 5.2% 5-20x, 1.8% 20-100x, 0.3% 100x+) ensuring statistical house edge while maintaining user engagement.
- Architected multi-database backends across client games utilizing both **SQL** and **NoSQL** engines (**MongoDB** and **Firebase Realtime Database / Firestore**) tailored to game mechanics.
- Collaborated with tier-1 international gaming clients (OneFC, Bombay Play) to deliver responsive PWA online gaming platforms and HTML5 titles with Redis and MongoDB caching.

Tech Stack: *TypeScript, Cocos Creator Engine, Node.js, RabbitMQ, MongoDB, Firebase, SQL, Redis, Angular, Flutter, Java, Kotlin, GCP*

KEY ENGINEERING & OPEN-SOURCE INITIATIVES

Zenith & Mitochondria — Autonomous Forensic Trading Workstation Infortts Open-Source Swarm

- Architected an autonomous quantitative trading engine integrated with MetaTrader 5, NLP market sentiment analysis, and vectorized trade analytics.
- Engineered sub-millisecond in-memory ring buffers and custom backtesting simulation engines streaming live telemetry via NATS JetStream to a Flutter workstation.

Tech Stack: *Python, C++, Go, NATS JetStream, PostgreSQL, Qdrant Vector DB, WebSockets, Flutter*

EDUCATION

Bachelor of Science (B.Sc.) in Computer Science — DDUC, University of Delhi 2018 – 2021 • New Delhi, India

SELECTED SPECIALIZATIONS

High-Frequency Trading (HFT) Architecture • Algorithmic Strategy & Quantitative Modeling • Distributed Multi-Agent Systems • Competitive Gaming (CS:GO, Chess)